Ministerul Educației și Cercetării al Republicii Moldova

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Facultatea Calculatoare, Informatică și Microelectronică

Laboratory Work 1:

Track A: The OS as a Resource Manager

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Lab 1 · Track A: The OS as a Resource Manager

System Identification

```
❯ whoami
ilie
❯ uname -a
Linux thinkbook 7.2.8-1-cachyos #1 SMP PREEMPT_DYNAMIC Sat, 26 Sep 2026 19:43:45 +0000 x86_64 GNU/Linux
❯ uptime
 16:09:20 up 23:50,  1 user,  load average: 3.28, 2.76, 2.18
```

Part 1. Files and directories (File management)

1. Navigation and Inspection

```
❯ pwd
/home/ilie/Documents/SO/os-lab1
❯ ls -la /
total 20
dr-xr-xr-x   1 root root  160 Aug 16 14:06 .
dr-xr-xr-x   1 root root  160 Aug 16 14:06 ..
lrwxrwxrwx   1 root root    7 Oct 12  2025 bin -> usr/bin
drwx------   5 root root 4096 Jan  1  1970 boot
drwxr-xr-x  20 root root 4360 Sep 27 16:18 dev
drwxr-xr-x   1 root root 4576 Sep 28 15:43 etc
drwxr-xr-x   1 root root   64 Aug  2 20:55 Games
drwxr-xr-x   1 root root    8 Jun 18 12:49 home
lrwxrwxrwx   1 root root    7 Oct 12  2025 lib -> usr/lib
lrwxrwxrwx   1 root root    7 Oct 12  2025 lib64 -> usr/lib
drwxr-xr-x   1 root root   14 Sep 13 23:30 mnt
drwxr-xr-x   1 root root  340 Sep 22 11:20 opt
drwxr-xr-x   1 root root    4 Aug 16 14:06 path
dr-xr-xr-x 389 root root    0 Sep 27 16:18 proc
drwxr-x---   1 root root  158 Aug 16 12:26 root
drwxr-xr-x  45 root root 1100 Sep 28 16:00 run
lrwxrwxrwx   1 root root    7 Oct 12  2025 sbin -> usr/bin
drwxr-x---   1 root root  302 Sep 28 15:48 .snapshots
drwxr-xr-x   1 root root   24 Aug 16 14:49 srv
dr-xr-xr-x  13 root root    0 Sep 27 16:18 sys
drwxrwxrwt  51 root root 1580 Sep 28 16:11 tmp
drwxr-xr-x   1 root root   94 Sep 28 15:43 usr
drwxr-xr-x   1 root root  152 Sep 27 16:18 var
❯ ls -la ~
total 22408
drwx------ 1 ilie ilie     1292 Sep 28 16:11 .
drwxr-xr-x 1 root root        8 Jun 18 12:49 ..
drwxr-x--- 1 ilie ilie       84 Sep 21 15:28 .android
drwxr-xr-x 1 ilie ilie       90 Jul 18 21:37 .ansible
drwxr-xr-x 1 ilie ilie       38 Jul 29 12:29 .antigravity-ide
drwxr-xr-x 1 ilie ilie      314 Sep  3 11:05 .anydesk
drwxr-x--- 1 ilie ilie        8 Aug 15 00:31 .atuin
-rw------- 1 ilie ilie      436 Aug  4 23:08 .bash_history
-rw-r--r-- 1 ilie ilie       21 Jun 18 12:49 .bash_logout
-rw-r--r-- 1 ilie ilie       57 Jun 18 12:49 .bash_profile
-rw-r--r-- 1 ilie ilie      202 Sep 21 15:20 .bashrc
drwxr-xr-x 1 ilie ilie       10 Sep 21 15:20 bin
-rwx------ 1 ilie ilie 22228992 Jun 21 12:14 blockchain.db
drwxr-xr-x 1 ilie ilie      806 Sep 27 21:46 Bottles
drwxr-xr-x 1 ilie ilie       14 Jul 18 12:50 .bun
drwx------ 1 ilie ilie     2362 Sep 28 16:09 .cache
drwxr-xr-x 1 ilie ilie      118 Aug  5 10:41 .cargo
drwxr-xr-x 1 ilie ilie     5006 Sep 28 15:41 .config
```

```
drwx------ 1 ilie ilie       36 Aug 13 12:48 .copilot
drwxr-xr-x 1 ilie ilie      316 Sep 27 21:27 Desktop
drwx------ 1 ilie ilie       66 Sep 14 12:20 .docker
drwxr-xr-x 1 ilie ilie      344 Sep 28 16:08 Documents
drwxr-xr-x 1 ilie ilie       12 Aug 10 18:22 .dotnet
drwxr-xr-x 1 ilie ilie     4386 Sep 28 15:55 Downloads
drwxr-xr-x 1 ilie ilie       10 Jun 18 15:52 .factorio
drwxr-xr-x 1 ilie ilie        6 Sep 27 19:17 Games
drwxr-xr-x 1 ilie ilie       42 Jul 29 12:29 .gemini
-rw-r--r-- 1 ilie ilie       80 Aug 18 17:02 .gitconfig
drwxr-xr-x 1 ilie ilie       30 Aug 20 12:15 .gitlab
drwx------ 1 ilie ilie      116 Jun 18 12:58 .gnupg
drwxr-xr-x 1 ilie ilie        6 Aug  5 10:40 go
-rw-r--r-- 1 ilie ilie      355 Sep 27 22:15 .gtkrc-2.0
drwxr-xr-x 1 ilie ilie      214 Jul 29 12:27 .kimi
drwx------ 1 ilie ilie       42 Aug 18 14:37 .kube
drwxr-xr-x 1 ilie ilie       10 Aug  5 12:39 .lemminx
drwxr-xr-x 1 ilie ilie       26 Aug  4 23:01 .local
drwxr-xr-x 1 ilie ilie       34 Aug  5 09:29 .m2
drwx------ 1 ilie ilie       44 Jun 22 00:07 .mozilla
drwxr-xr-x 1 ilie ilie        0 Jun 18 12:52 Music
drwxr-xr-x 1 ilie ilie       92 Sep 14 11:46 .npm
drwxr-xr-x 1 ilie ilie       66 Aug  5 10:49 .nvm
-rw-r--r-- 1 ilie ilie     9886 Jul  8 22:32 .p10k.zsh
drwxr-xr-x 1 ilie ilie       36 Aug 25 14:37 Pictures
drwx------ 1 ilie ilie       10 Jun 18 13:04 .pki
drwxr-xr-x 1 ilie ilie      122 Sep 27 16:35 Projects
drwxr-xr-x 1 ilie ilie        0 Jun 18 12:52 Public
-rw------- 1 ilie ilie      256 Jun 18 14:28 .pulse-cookie
-rw-r--r-- 1 ilie ilie        0 Aug  4 23:04 .python_history
drwxr-xr-x 1 ilie ilie        6 Aug  9 11:47 .rancher
drwxr-xr-x 1 ilie ilie       78 Aug 14 15:13 .renpy
-rw-r--r-- 1 ilie ilie      317 Sep 28 14:00 sodomized.cow
drwxr-xr-x 1 ilie ilie      186 Sep  9 16:47 .ssh
drwxr-xr-x 1 ilie ilie      190 Sep 17 12:06 .steam
lrwxrwxrwx 1 ilie ilie       29 Sep 14 02:03 .steampath -> /home/ilie/.steam/sdk32/steam
lrwxrwxrwx 1 ilie ilie       27 Sep 14 02:03 .steampid -> /home/ilie/.steam/steam.pid
drwxr-xr-x 1 ilie ilie       24 Aug 10 18:23 .sts4
drwxr-xr-x 1 ilie ilie        0 Jun 18 12:52 Templates
drwxr-xr-x 1 ilie ilie       72 Jul 22 15:53 .terraform.d
drwxr-xr-x 1 ilie ilie        6 Jun 18 12:52 .var
-rw------- 1 ilie ilie       10 Jul 19 20:35 .vault_pass
drwxr-xr-x 1 ilie ilie       14 Jul 23 11:27 Videos
-rw------- 1 ilie ilie    16189 Jul 22 16:22 .viminfo
drwxr-xr-x 1 ilie ilie       44 Aug 10 18:20 .vscode
drwxr-xr-x 1 ilie ilie       26 Aug 10 18:20 .vscode-shared
-rw-r--r-- 1 ilie ilie      219 Aug 16 13:24 .wget-hsts
drwxr-xr-x 1 ilie ilie      126 Sep 27 16:45 .wine
-rw-r--r-- 1 ilie ilie       50 Aug 24 01:10 .Xresources
-rw-r--r-- 1 ilie ilie    57149 Jul 14 02:06 .zcompdump-thinkbook-5.9.1
-r--r--r-- 1 ilie ilie   123024 Jul 14 02:06 .zcompdump-thinkbook-5.9.1.zwc
-rw-r--r-- 1 ilie ilie    57469 Sep 27 19:21 .zcompdump-thinkbook-5.9.2
-r--r--r-- 1 ilie ilie   124872 Sep 27 19:21 .zcompdump-thinkbook-5.9.2.zwc
-rw-r--r-- 1 ilie ilie       47 Aug  4 23:02 .zshenv
-rw------- 1 ilie ilie   251856 Sep 28 16:11 .zsh_history
-rw-r--r-- 1 ilie ilie      706 Aug 15 01:41 .zshrc
❯ cd /etc && ls | head
adjtime
alfis.toml
alsa
ananicy.d
arch-release
arptables.conf
audisp
audit
```

```
avahi
bash.bash_logout
```

2. Create, Copy, Rename, and Remove

```
❯ cd /home/ilie/Documents/SO/os-lab1
❯ mkdir -p demo && cd demo
❯ echo "hello operating systems" > note.txt
❯ cp note.txt copy.txt
❯ mv copy.txt renamed.txt
❯ ls -l
total 8
-rw-r--r-- 1 ilie ilie 24 Sep 28 16:22 note.txt
-rw-r--r-- 1 ilie ilie 24 Sep 28 16:22 renamed.txt
❯ rm renamed.txt
❯ ls -l
total 4
-rw-r--r-- 1 ilie ilie 24 Sep 28 16:22 note.txt
```

3. Ownership and Permissions

```
❯ ls -l note.txt
-rw-r--r-- 1 ilie ilie 24 Sep 28 16:22 note.txt
❯ chmod 600 note.txt
❯ ls -l note.txt
-rw------- 1 ilie ilie 24 Sep 28 16:22 note.txt
❯ chmod 644 note.txt
❯ ls -l note.txt
-rw-r--r-- 1 ilie ilie 24 Sep 28 16:22 note.txt
```

Observe (Part 1 Answers)

1. **Who owns the files you just created, and which group?**
   - **Owner:** `ilie` (user ID `1000`).
   - **Group:** `ilie` (group ID `1000`).
   - The OS automatically assigned the credentials of the active user session as the owner and primary group of the newly created files.
2. **What do the ten characters at the start of `ls -l` mean for `note.txt` after `chmod 600`?**
   - The string is `-rw-------`.
     - `1st character (-)`: File type indicator, showing this is a standard regular file (not a directory `d` or symbolic link `l`).
     - `Characters 2–4 (rw-)`: User/Owner permissions. Read (`r`) and write (`w`) permissions are granted, while execute (`-`) is denied.
     - `Characters 5–7 (---)`: Group permissions. Read, write, and execute permissions are all denied.
     - `Characters 8–10 (---)`: Others/World permissions. Read, write, and execute permissions are all denied.
   - In octal, `600` represents binary `110 000 000`, restricting access exclusively to the owner.
3. **Name two directories directly under `/` and say, in one line each, what they hold.**
   - `/etc`: Contains system-wide configuration files and initialization scripts for installed packages and services.
   - `/proc`: A virtual pseudo-filesystem synthesized dynamically by the Linux kernel that exposes runtime process and kernel hardware state.

Interesting Output Highlight (Part 1)

Inspecting `ls -la /` reveals that virtual filesystem mount points like `/proc` and `/sys` report a size of `0` bytes. Unlike real directories on the storage drive, they consume no disk space because they are dynamically populated in-memory by the kernel on demand.

Part 2. Processes (Process management)

4. Process Listing

```
❯ ps aux | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.0  23292 15712 ?        Ss   Sep27   0:19 /usr/lib/systemd/systemd --switched-root
↪  --system --deserialize=60 splash
root           2  0.0  0.0      0     0 ?        S    Sep27   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    Sep27   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   Sep27   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   Sep27   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   Sep27   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<   Sep27   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   Sep27   0:00 [kworker/R-netns]
root          11  0.0  0.0      0     0 ?        I<   Sep27   0:00 [kworker/0:0H-kblockd]

❯ ps aux | wc -l
377
```

5. Managing Background Jobs

```
❯ sleep 300 &
[1] 599116

❯ jobs
[1]+  Running                 sleep 300 &

❯ ps -ef | grep sleep
ilie      599116  584870  0 16:22 pts/5    00:00:00 sleep 300
ilie      599118  584870  0 16:22 pts/5    00:00:00 grep sleep

❯ kill 599116
[1]+  Terminated              sleep 300
```

6. Process Introspection via `/proc`

```
❯ sleep 300 &
[1] 599119

❯ PID=599119

❯ ls /proc/$PID/
arch_status         fdinfo              ns              smaps_rollup
attr                gid_map             numa_maps       stack
autogroup           io                  oom_adj         stat
auxv                ksm_merging_pages   oom_score       statm
cgroup              ksm_stat            oom_score_adj   status
clear_refs          limits              pagemap         syscall
cmdline             loginuid            personality     task
comm                map_files           projid_map      timens_offsets
coredump_filter     maps                root            timers
cpu_resctrl_groups  mem                 sched           timerslack_ns
cwd                 mountinfo           schedstat       uid_map
environ             mounts              sessionid       wchan
exe                 mountstats          setgroups
fd                  net                 smaps

❯ cat /proc/$PID/status | head -20
Name:   sleep
Umask:  0022
State:  S (sleeping)
Tgid:   599119
Ngid:   0
Pid:    599119
PPid:   584870
```

```
TracerPid:  0
Uid:    1000    1000    1000    1000
Gid:    1000    1000    1000    1000
FDSize: 256
Groups: 3 90 954 959 980 982 983 985 989 992 995 998 1000
NStgid: 599119
NSpid:  599119
NSpgid: 584870
NSsid:  584870
Kthread:    0
VmPeak:     6312 kB
VmSize:     6312 kB
VmLck:         0 kB
```

```
❯ kill $PID
```

Observe (Part 2 Answers)

1. **What is the PID of process number 1, and what is it (look it up)?**
   - **PID:** 1.
   - **Program:** `systemd` (`/usr/lib/systemd/systemd`).
   - It is the init system and service manager for Linux, acting as the very first user-space process started by the Linux kernel after booting; all other user processes and background service daemons are direct or indirect descendants of process 1.
2. **Roughly how many processes were running on your idle VM?**
   - About `375` processes (`ps aux | wc -l`). A large portion of these are internal kernel threads (such as `[kthreadd]`, `[kworker/...]`, and interrupt handlers) that assist the OS in managing hardware, alongside systemd background daemons and desktop environment tasks.
3. **From `/proc/<pid>/status`, what does the "State:" line say for your sleeping process?**
   - `State: S (sleeping)`. This designates an interruptible sleep state, indicating the process is idle and waiting on an OS timer event (the expiration of its 300-second sleep interval) without consuming CPU cycles.

Interesting Output Highlight (Part 2)

The contents of `/proc/$PID/` showcase how the OS kernel exposes internal process control blocks (PCBs) as standard directories. Key entries like `exe` (symlink to the binary), `fd/` (directory of active file descriptor handles), and `maps` (virtual address space layout) allow direct inspection of a running program without attached debuggers.

Part 3. Memory (Memory management)

Memory Inspection

```
❯ free -h
               total        used        free      shared  buff/cache   available
Mem:            19Gi       9.3Gi       5.8Gi       191Mi       4.9Gi        10Gi
Swap:           19Gi       2.1Gi        17Gi
```

```
❯ cat /proc/meminfo | head -6
MemTotal:       20338228 kB
MemFree:         6081672 kB
MemAvailable:   10608164 kB
Buffers:            2248 kB
Cached:          4431108 kB
SwapCached:          712 kB
```

```
❯ sleep 300 & PID=$!
[1] 599201
```

```
❯ grep VmRSS /proc/$PID/status
```

```
VmRSS:      1816 kB

❯ kill $PID
```

Observe (Part 3 Answers)

1. **How much total RAM does the VM have, and how much is free?**
   - **Total RAM:** Approximately `19 GiB` (`20,338,228 kB` ~ 20.3 GB).
   - **Free RAM:** `5.8 GiB` strictly unallocated (`MemFree`), while `10 GiB` is reported as `available` (`MemAvailable`). The `available` value reflects actual usable memory because the OS can immediately evict reclaimable page caches and file buffers (`buff/cache: ~4.9 GiB`) if application programs request memory.
2. **What is swap, and how much is configured?**
   - **What swap is:** Swap is an overflow area (either a dedicated disk partition, file, or compressed RAM segment) utilized by the OS virtual memory manager to page out inactive memory pages, freeing physical RAM for active working sets and caching.
   - **Configured swap:** `19 GiB` (specifically `19.4 GiB` configured via `/dev/zram0`, an in-RAM compressed block device), with `2.1 GiB` used and `17 GiB` free.
3. **How much resident memory (VmRSS) does a bare `sleep` process use, and does that surprise you?**
   - **Resident memory:** `1816 kB` (~1.8 MB).
   - **Reflection:** While 1.8 MB initially seems large for a program that merely pauses execution, it is expected under modern Linux. The resident memory accounts not only for the minimal code of the `sleep` binary itself, but also for mapping shared C libraries (`libc.so`), the runtime dynamic linker (`ld-linux.so`), the stack, and kernel page table structures allocated to manage the process's virtual memory space.

Interesting Output Highlight (Part 3)

The comparison between `MemFree: 6081672 kB` (~6.08 GB) and `MemAvailable: 10608164 kB` (~10.6 GB) in `/proc/meminfo` demonstrates the efficiency of the kernel page cache. The OS keeps ~4.4 GB of previously read disk pages cached in RAM rather than letting the memory sit unused, accelerating future I/O while still guaranteeing that this RAM can be reclaimed instantly when needed.

Part 4. Devices and storage (I/O management)

Storage and Device Listing

```
❯ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme0n1p2  235G  145G   90G  62% /
devtmpfs        9.7G     0  9.7G   0% /dev
tmpfs           9.7G   56M  9.7G   1% /dev/shm
efivarfs        148K  140K  3.4K  98% /sys/firmware/efi/efivars
tmpfs           3.9G  2.1M  3.9G   1% /run
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
/dev/nvme0n1p2  235G  145G   90G  62% /home
/dev/nvme0n1p2  235G  145G   90G  62% /root
/dev/nvme0n1p2  235G  145G   90G  62% /srv
/dev/nvme0n1p2  235G  145G   90G  62% /var/cache
tmpfs           9.7G  1.2G  8.6G  12% /tmp
/dev/nvme0n1p2  235G  145G   90G  62% /var/tmp
/dev/nvme0n1p2  235G  145G   90G  62% /var/log
/dev/nvme0n1p1  4.0G  286M  3.8G   7% /boot
/dev/nvme1n1p1  469G  259G  186G  59% /mnt/storage
tmpfs           2.0G  118M  1.9G   6% /run/user/1000

❯ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
zram0       253:0    0  19.4G  0 disk [SWAP]
nvme0n1     259:0    0 238.5G  0 disk
```

```
├─nvme0n1p1 259:1    0     4G  0 part /boot
└─nvme0n1p2 259:2    0 234.5G  0 part /var/log
                                      /var/tmp
                                      /var/cache
                                      /srv
                                      /root
                                      /home
                                      /
nvme1n1     259:3    0 476.9G  0 disk
└─nvme1n1p1 259:4    0 476.9G  0 part /mnt/storage
```

```
❯ du -sh /home/ilie/Documents/SO/os-lab1
12K /home/ilie/Documents/SO/os-lab1
```

```
❯ ls -l /dev | head
total 0
crw-r--r--   1 root root    10,   235 Sep 28 15:43 autofs
drwxr-xr-x   2 root root          160 Sep 28 15:43 block
crw-rw----   1 root disk    10,   234 Sep 28 15:43 btrfs-control
drwxr-xr-x   3 root root           60 Sep 27 16:18 bus
drwxr-xr-x   2 root root         4540 Sep 28 15:48 char
crw-------   1 root root     5,     1 Sep 28 15:43 console
lrwxrwxrwx   1 root root           11 Sep 27 16:18 core -> /proc/kcore
drwxr-xr-x   6 root root          120 Sep 28 15:27 cpu
crw-rw----   1 root audio   10,   259 Sep 28 15:43 cpu_dma_latency
```

```
❯ mount | head
/dev/nvme0n1p2 on / type btrfs
 ↪  (rw,noatime,compress=zstd:1,ssd,discard=async,space_cache=v2,subvolid=256,subvol=/@)
devtmpfs on /dev type devtmpfs (rw,nosuid,size=10078308k,nr_inodes=2519577,mode=755,inode64,huge=advise)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,inode64,huge=advise,usrquota)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=600,ptmxmode=000)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime)
none on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime)
efivarfs on /sys/firmware/efi/efivars type efivarfs (rw,nosuid,nodev,noexec,relatime)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
```

Observe (Part 4 Answers)

1. **Which device is your root filesystem "/" mounted on?**
   - It is mounted on `/dev/nvme0n1p2` (partition 2 of the primary NVMe solid-state drive `nvme0n1`, using the `btrfs` filesystem with zstd compression).
2. **Give one entry from `/dev` and say what real thing it stands for.**
   - `/dev/nvme0n1` represents the physical NVMe solid-state drive (a block storage hardware device), while `/dev/console` represents the primary system console character device used to handle system output and interactive input.
3. **In one sentence: what does "everything is a file" mean, based on what you saw?**
   - "Everything is a file" means that the operating system abstracts diverse physical hardware devices, inter-process communication channels, and kernel runtime state through a single unified filesystem namespace, enabling programs to access storage disks, memory data, and peripheral devices using uniform file operations such as `open`, `read`, and `write`.

Interesting Output Highlight (Part 4)

In `ls -l /dev`, instead of displaying regular file sizes in bytes, entries display two numbers separated by a comma (for example, `10, 235` for `autofs` or `5, 1` for `console`). These correspond to the **major and minor device numbers**, where the major number identifies the kernel driver assigned to manage the device, and the minor number specifies the unique physical or logical instance of that device.

Wrap-up and Deliverable: Closing Synthesis

The operating system acts as the foundational resource manager of a computer by coordinating four core hardware resources: files and storage, processes, physical memory, and I/O devices.

We observed file management through commands like `ls -l` and `chmod`, which revealed how the OS translates raw storage into a structured tree governed by access permissions, while process management was demonstrated using `ps` and `/proc/<pid>/status` as the kernel created, isolated, scheduled, and tracked running tasks.

Finally, memory management was made visible with `free -h` and `meminfo` as the kernel balanced physical RAM allocations against dynamic caches and swap, while `lsblk` and `/dev` showed device management in action by exposing disparate hardware components through uniform file-like interfaces.